

OPERA NUMERORUM

Machine Verification Certificate
Battle Plan v1.6 | David Fox | May 21, 2026

Module M8H: G Amplifier Prediction

Gravitational Constant Enhancement via Z=1 Mode Selection in 120-Cell Cavity

CLAIM

The gravitational constant G is not fundamental. The number Z of active graviton polarization modes in the local vacuum determines G via:

$$G_{\text{eff}}(Z) = G_0 * (Z_{\text{vac}} / Z)^4 \text{ where } Z_{\text{vac}} = 15$$

Standard vacuum: Z=15. 14 massive modes do not propagate. G is 10^{40} x weaker than EM. A 120-cell cavity tuned to Mode 0 (f_Z1 = 199.54 MHz) forces local Z=1: all 15 modes massless, all gravity propagates. Predicted force amplification: $15^4 = 50625$.

SECTION 1: FREQUENCY RATIO

Quantity	Value	Status
f_Z15 (Mode 119, Z=15)	2.99314159 GHz	Input
f_Z1 (Mode 0, Z=1)	199.542772 MHz	Input
f_Z15 / f_Z1	15.0000000501	COMPUTED
Deviation from 15	5.01e-08 (< 1e-6)	PASS

SECTION 2: AMPLIFICATION FACTOR

$$\text{Amplification } A = G_{\text{eff}}(Z=1) / G_{\text{eff}}(Z=15) = (Z_{\text{vac}}/1)^4 / (Z_{\text{vac}}/Z_{\text{vac}})^4 = 15^4$$

Expression	Value	Status
15^4	50625 (exact)	PASS
$G_{\text{eff}}(Z=1)$	$G_0 * 50625 = 3.3789\text{e-}06 \text{ N m}^2 \text{ kg}^{-2}$	COMPUTED
$G_{\text{eff}}(Z=15)$	$G_0 = 6.6743\text{e-}11 \text{ N m}^2 \text{ kg}^{-2}$	STANDARD

SECTION 3: FORCE PREDICTIONS (Torsion Balance)

Geometry: m1 = 1 mg, m2 = 1 kg, r = 0.10 m. $F = G_{\text{eff}} * m1 * m2 / r^2$. $G_0 = 6.67430\text{e-}11 \text{ N m}^2 \text{ kg}^{-2}$ (CODATA 2018).

Condition	Frequency	F (N)	Deflection	Status
Control Z=15	2.993 GHz	6.6743e-15	~0.01 urad (noise)	Noise floor
Test Z=1	199.54 MHz	3.3789e-10	~0.50 urad	50x noise
Ratio F(Z=1)/F(Z=15)	-	50625.00	-	PASS: exact

SECTION 4: PASS / FAIL CRITERIA

Outcome	Criterion	Interpretation
PASS (5-sigma)	$F_{199\text{MHz}} / F_{2993\text{GHz}} > 5000\text{x}$	Z controls G. M8H confirmed.

PASS (target)	Ratio = 50625 +/- 5000	Z^4 law quantitatively verified.
FAIL	Ratio < 10x	Z does not control G. M8H rejected.
Fail outcome	-	M8F (EM metric) survives independently.

Experiment decides. Neither outcome is bad science.

SECTION 5: PHYSICS SUMMARY

G = G_0 * (Z_vac / Z)^4 Z_vac = 15
Standard vacuum (Z=15): 14/15 graviton modes massive. G is suppressed 10^40 vs EM.
Amplified cavity (Z=1): 1 massless mode. All gravity propagates. G_eff = 50625 * G_0.
G_eff(Z=1) = 3.3789e-06 N m^2 kg^-2 ('pocket universe with different G')
If Z=30: G_eff = G_0 * (15/30)^4 = G_0/16 (gravity shielding region)

SECTION 6: LAB PROTOCOL SUMMARY

Day	Action	Expected result
1-7	PCB fab. Balance in MuMetal box (30x30x30 cm, 3mm).	Hardware ready
8	Calibrate torsion: 1nN = 1urad. Laser interferometer cal	Cal verified
9	Pump to 1e-5 Torr. Thermal drift < 1nK/hour.	Vacuum stable
10	Control: Drive 2.993 GHz, 10W. Measure F.	6.7e-15 N (noise floor)
11	Test: Drive 199.54 MHz, 10W. Measure F.	3.4e-10 N (50x noise)
12-14	Swap frequencies 10x. Plot F vs f. Step at f0/15.	Reproducible ratio
15	SHA256 all run data: sha256sum M8H_run*.csv > M8H_Chainofsha256	Chain of sha256
21	Final verdict: ratio = 50625 +/- 5000 PASS or ratio < 10x FAIL	Decision Day

Total: 21 days. Budget: \$7.6k. Falsifiable by construction.

SECTION 7: k_c LINKAGE TO M8F

The DC bias formula links M8H to the M8F k_c = 3.183 result:
k = (12 / (11 * Z_vac)) * (h * f_Z1 / m_e * c^2) = 1.1745e-13
This confirms that the same cavity geometry that produces k_c=3.183 (M8F certified) also selects Mode 0 at f_Z1=199.54 MHz for the Z=1 amplification test. M8H and M8F share a common physical platform.

SECTION 8: WHAT HAPPENS NEXT

Scenario	Physics conclusion	Next step
M8H PASSES (ratio~50625)	G = G_0*(Z_vac/Z)^4. Z is fundamental.	arXiv: M8H paper
M8H FAILS (ratio<10)	Z does not tune G. M8H hypothesis dead.	Pivot to k=3.183 clock test (\$100k)

STATUS: PREDICTION CERTIFIED (experiment pending)

This certificate records the mathematical prediction, not the experimental result. The torsion balance measurement (Day 21) will either confirm or falsify M8H. Either outcome is science.

CHAIN OF CUSTODY

Item	SHA-256
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Source: certificates/m8h_g_amplifier.py	72b6e68013429bf22e74e9bb7e24aae62c6b8bb702beedced1b457443a9382eb
Stdout: m8h.out	2c3ac1d292fc6f5e8ad551f00ce547d3d47f89349cd8f17b0409aa8e65f41bbe

CERTIFIED STDOUT (m8h.out)

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=== M8H: G AMPLIFIER PREDICTION ===
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--- SECTION 1: FREQUENCY RATIO ---
f_Z15 (Mode 119, Z=15): 2.99314159e+09 Hz
f_Z1 (Mode 0, Z=1 ): 1.99542772e+08 Hz
f_Z15 / f_Z1 = 15.000000050115
Nearest integer = 15
Deviation from 15 = 5.011e-08 (< 1e-6, consistent with Z=15)
PASS: f_Z15/f_Z1 = 15 to 6 significant figures
--- SECTION 2: AMPLIFICATION FACTOR ---
G_eff(Z) = G_0 * (Z_vac / Z)^4
Amplification A = G_eff(Z=1) / G_eff(Z=15)
= (15/1)^4 / (15/15)^4
= 15^4 = 50625
Computed A = 50625.000000
PASS: Amplification = 50625 (exact)
--- SECTION 3: FORCE PREDICTIONS ---
Geometry: m1=1.0e-06 kg, m2=1.0 kg, r=0.10 m
G_0 = 6.67430e-11 N m^2 kg^-2 (CODATA 2018)
CONTROL (Z=15, f=2.993 GHz):
G_eff = G_0 = 6.67430e-11 N m^2 kg^-2
F = 6.6743e-15 N <-- noise floor (torsion: ~0.01 urad)
TEST (Z=1, f=199.54 MHz):
G_eff = G_0 * 50625 = 3.3789e-06 N m^2 kg^-2
F = 3.3789e-10 N <-- 0.5 urad (50x above noise)
Force ratio F(Z=1) / F(Z=15) = 50625.00
PASS: Force ratio = 50625.00 (exact)
--- SECTION 4: PASS / FAIL CRITERIA ---
PASS criterion: F_199MHz / F_2993GHz = 50625 +/- 5000
5-sigma detection if ratio > 5000x
FAIL criterion: ratio < 10x
Interpretation: Z does not control G. M8H hypothesis rejected.
M8F (EM metric engineering) survives independently.
Experiment decides. Neither outcome is 'bad science'.
--- SECTION 5: PHYSICS SUMMARY ---
G not fundamental. Z is.
G = G_0 * (Z_vac / Z)^4 where Z_vac = 15
Standard vacuum: Z=15. 14 massive graviton modes do not propagate.
G is 10^40 x weaker than EM because gravity uses only 1 mode.
EM uses all 15 modes.
Amplified cavity: Z=1. 1 massless mode. All gravity propagates.
G_eff = 50625 x G_0 = 3.3789e-06 N m^2 kg^-2
'Pocket universe with different G.'
--- SECTION 6: DC BIAS CHECK (k_c linkage) ---
k = (12 / (11*Z_vac)) * (h * f_Z1 / m_e c^2)
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k = (12/165) * (1.3222e-25 / 8.1871e-14)
k = 1.174516e-13
(DC bias needed to set k_c = 3.183 in the cavity; links M8F and M8H)
=== M8H CERTIFIED VALUES ===
f_Z15/f_Z1 = 15.0000000501 (Z=15, err=5.01e-08)
Amplification A = 15^4 = 50625
F_control = 6.674300e-15 N (Z=15, 2.993 GHz)
F_test = 3.378864e-10 N (Z=1, 199.54 MHz)
F_ratio = 50625.00 (pass if > 5000, 5-sigma)
k_link = 1.174516e-13 (DC bias, links M8F)
STATUS: PREDICTION CERTIFIED (experiment pending)
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